

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, ML Evaluation Aug 2025 to May 2026

- Ran **evals** on RAG workflows by comparing chunking strategies, embedding models, and top-k settings, and classified each failure as retrieval-caused or generation-caused.
- Boosted malware-family classification to **80.2%** accuracy across **17 families** by soft-voting **ResNet50**, **ViT-B/16**, VGG16, HOG, and Haralick models, beating the best single model by **2.4 pts** (McNemar $p < 0.001$).
- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.

Turito | Software Engineering Intern, Systems Performance Jun 2022 to Oct 2022

- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Prototyped a **Rust** + **QUIC** media path that supported about **8,000** lightweight subscribers with about **24%** lower startup time.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Reduced performance-investigation time by about **40%** with tooling built on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, Rust, C++17, TypeScript, SQL

Evals & LLMs: agent evaluation, RAG evals, LLM-as-judge, statistical testing, OpenAI and Anthropic APIs, garak

ML: PyTorch, Hugging Face, ViT, ensembles, embeddings

Infra: Temporal, Kafka, gRPC, FastAPI, Docker, PostgreSQL, Redis, Qdrant

PROJECTS

Teller, Agent Evaluation Environment | *Browser agents, Python, TypeScript*

- Built an adversarial target app (nested frames, missing test IDs, session expiry, permission denials) and analyzed **14 agent runs**, classifying failures as planning, execution, validation, or recovery.
- Designed a capability layer so the same workflows run on a second tenant with a **16-line** overlay and no re-recording.

AutoGrade AI, Multi-Agent Evaluation Platform | *Python, Temporal, Kafka, gRPC, PostgreSQL, Qdrant*

- Built rubric-driven grading with multiple LLM agents, repeated evaluation to expose disagreement, confidence gates, human-review queues, and audit traces over **Temporal** workflows.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Distributed Computing, Database Systems, Advanced Algorithms, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)